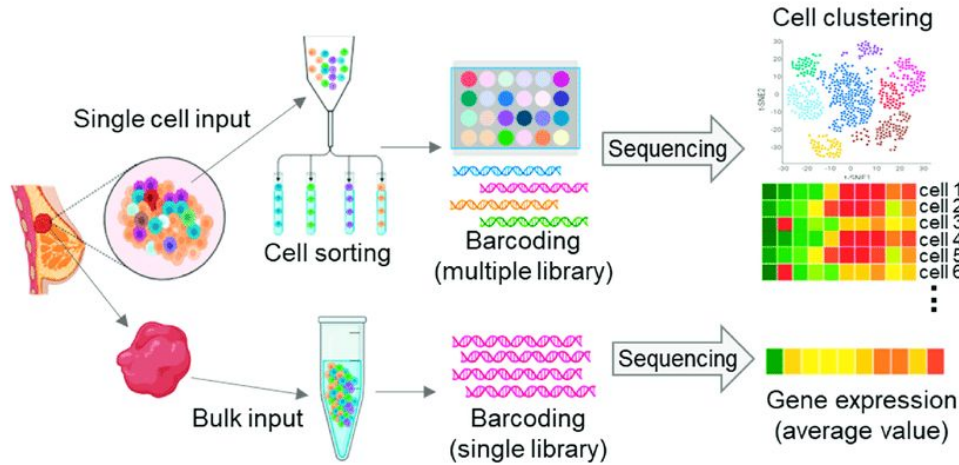


ANÁLISIS DE MÉTODOS DE EXPRESIÓN DIFERENCIAL CON DATOS DE scRNA-seq

Maria Alejandra Rojo Orozco
Johan Hernando Largo Gonzalez
Jenny Lorena Patiño Rosas

Introducción



La tecnología de secuenciación de única célula a tomado relevancia al tener ventajas con respecto a la secuenciación de RNA-seq convencional (Svensson, V. et al., 2020).

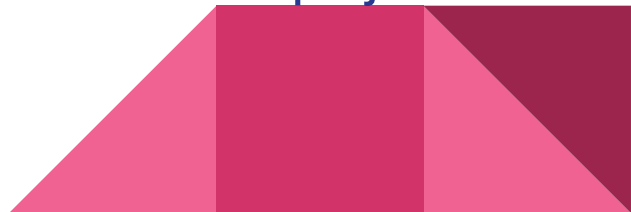
- Mejor comprensión de la heterogeneidad del tejido (Cao, J. et al., 2020).
- Caracterizaciones de poblaciones de células raras (Jindal, A. et al., 2018; Nguyen, A. et al., 2018).
- Etiología de la enfermedad impulsada por el tipo celular (Schirmer, L. et al., 2019; Reyfman, P. A. et al., 2019)

Pregunta biológica

Oportunidad: Identificación de genes expresados diferencialmente entre grupos, mediante el uso de métodos apropiados.

Considerando el desafío que implica el análisis de datos del scRNA-seq, asociado a aspectos como eliminación de genes poco expresados, la escasez de conjuntos de datos de scRNA-seq, variación de célula a célula dentro de un sujeto, entre otros, nos planteamos la siguiente pregunta de investigación:

¿Cuál es la desempeño de los métodos estadísticos *T-test, DESeq Y MAST* para el análisis de expresión diferencial en datos simulados de scRNA-seq planteados en diferentes niveles de complejidad?



Objetivo

Evaluar el rendimiento de los métodos estadísticos *prueba t*, *DESeq2* y *MAST* en el análisis de expresión diferencial usando datos de Single Cell Sequencing.

Objetivos Específicos

- Generar datos simulados de prueba con la biblioteca splatter.
 - Ejecutar los diferentes métodos de expresión diferencial con los datos generados.
 - Comparar los resultados obtenidos
 - Realizar recomendaciones para el análisis de expresión diferencial en datos de célula única.
- 

Diseño Experimental

Obtención y procesamiento
de datos

Ejecución de métodos de
DE para datos de
scRNA-seq

Comparación de
los DEGs

Conclusiones

**Simulación de datos con
la librería splatter para
facilitar la identificación
de grupos**

**T-test
DESeq
MAST**



Obtención y Preprocesamiento de los datos

Datos de referencia :

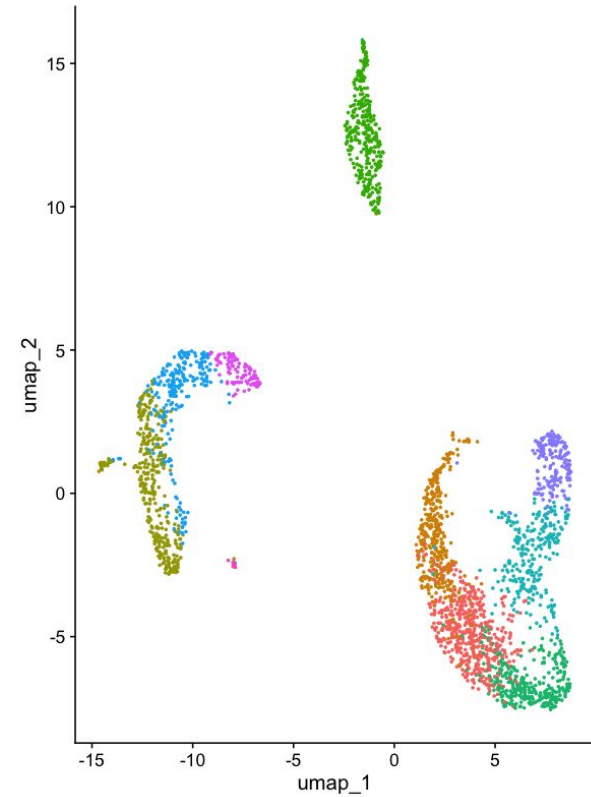
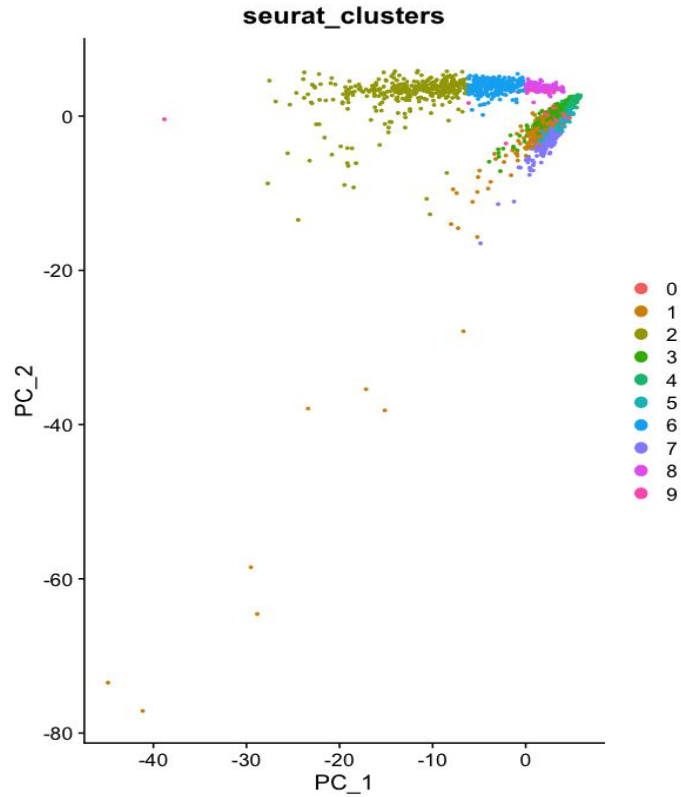
```
> library("Seurat") #single cell pre-processing and processing PBMC Data
> library("SeuratData") #data reference PBMC
> pbmc3k
An object of class Seurat
13714 features across 2700 samples within 1 assay
Active assay: RNA (13714 features, 0 variable features)
2 layers present: counts, data
```

Tipo de células: Células mononucleares de sangre periférica (PBMC)

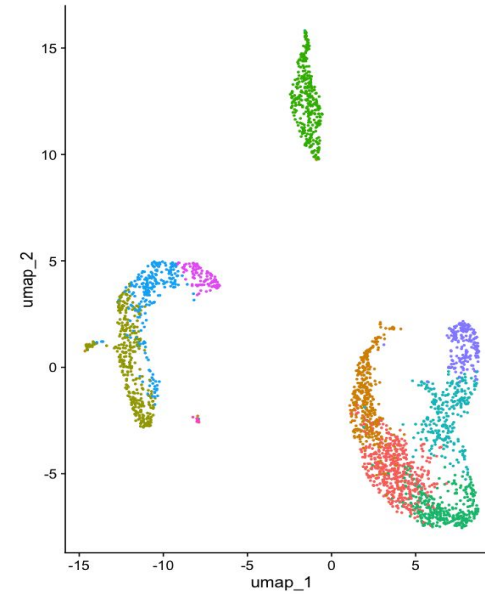
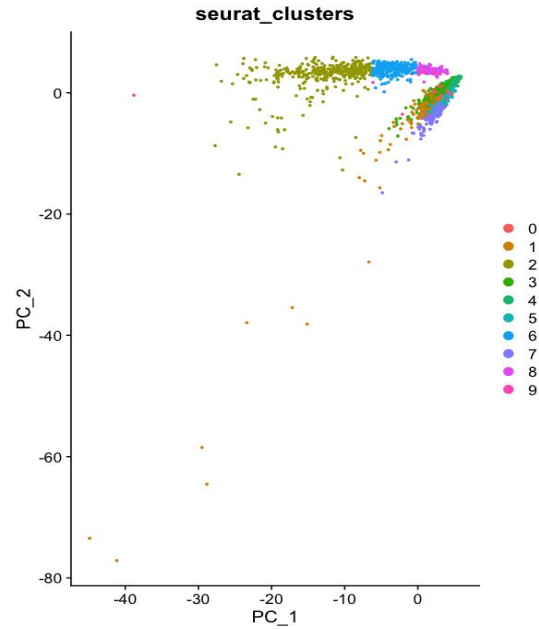
```
> tab1 <- round(table(data$seurat_annotatons)/sum(table(data$seurat_annotatons)),2)
> tab1
```

Naive	CD4 T	Memory	CD4 T	CD14+ Mono	B	CD8 T	FCGR3A+ Mono
	0.26		0.18	0.18	0.13	0.10	0.06
	NK		DC	Platelet			
	0.06		0.01	0.01			

DATOS DE REFERENCIA



DATOS DE REFERENCIA



Naive CD4 T	Memory CD4 T	CD14+ Mono	B	CD8 T	FCGR3A+ Mono
0.26	0.18	0.18	0.13	0.10	0.06
NK	DC	Platelet			
0.06	0.01	0.01			

Simulación 1

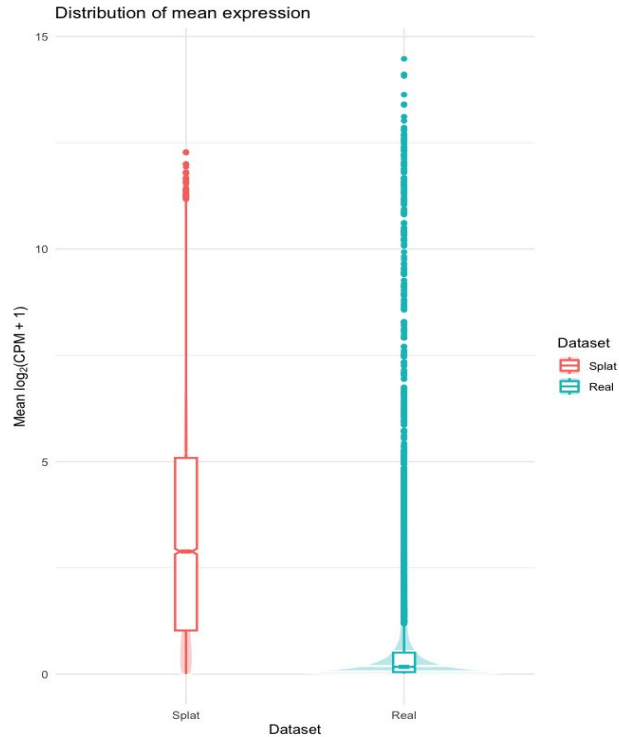
```
sim.groups <- splatSimulateGroups(batchCells=3000, #number of cells per batch
  group.prob = c(0.55, 0.38, 0.04, 0.01, 0.02), # Sim 1
  de.prob = c(0.08, 0.06, 0.05, 0.05, 0.01), #probability DEGS per group
  de.facLoc = c(0.01, 0.2, 0.1, 0.1, 0.1), #define more or less extreme differences between groups
  de.facScale = c(0.2, 0.5, 0.2, 0.5, 0.4), #define more or less extreme differences between groups
  de.downProb = 0, #probability of subexpressed genes
  sparsify = TRUE, #a lot of zeros
  verbose = TRUE) #print additional info
```

Simulación 2

```
sim.groups_2 <- splatSimulateGroups(batchCells=3000, #number of cells per batch
  group.prob = c(0.20, 0.20, 0.20, 0.20, 0.20), # Sim 2 #number and proportion groups
  de.prob = c(0.08, 0.06, 0.05, 0.05, 0.01), # probability DEGS per group
  de.facLoc = c(0.01, 0.2, 0.1, 0.1, 0.2), #define more or less extreme differences between groups
  de.facScale = c(0.8, 0.8, 0.8, 0.8, 0.8), #define more or less extreme differences between groups
  downProb = 0, #probability of subexpressed genes
  sparsify = TRUE, #a lot of zeros
  verbose = TRUE) #print additional info
```

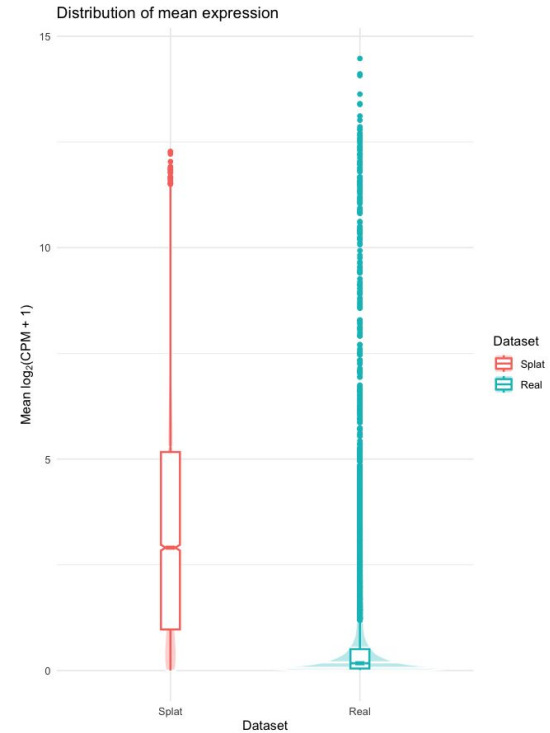
Simulación 1

```
group.prob = c(0.55, 0.32, 0.06, 0.05, 0.02),
```



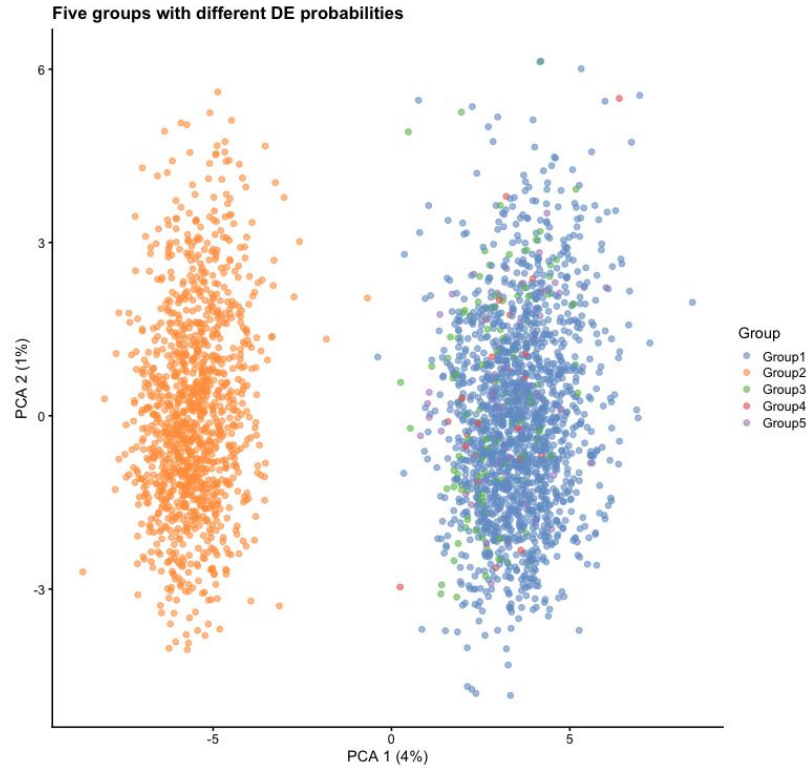
Simulación 2

```
group.prob = c(0.20, 0.20, 0.20, 0.20, 0.20),
```



PCA Inicial

Simulación 1



Simulación 2

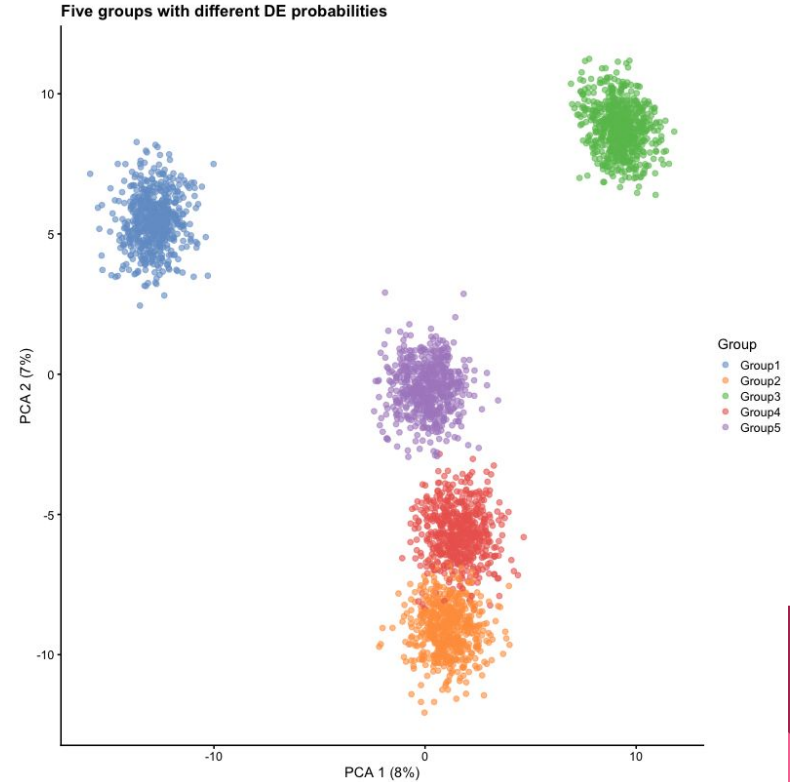
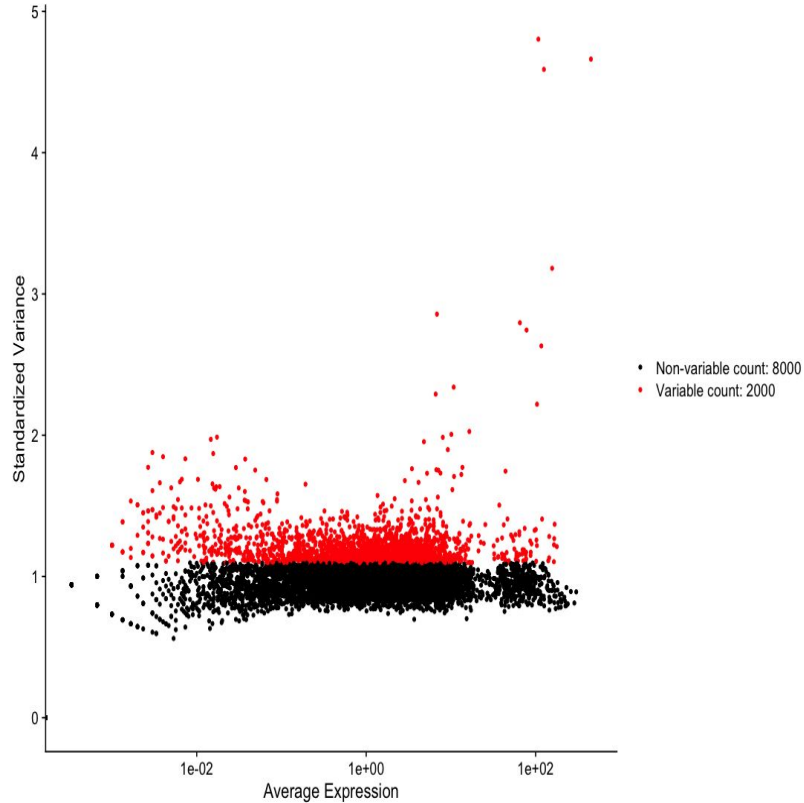


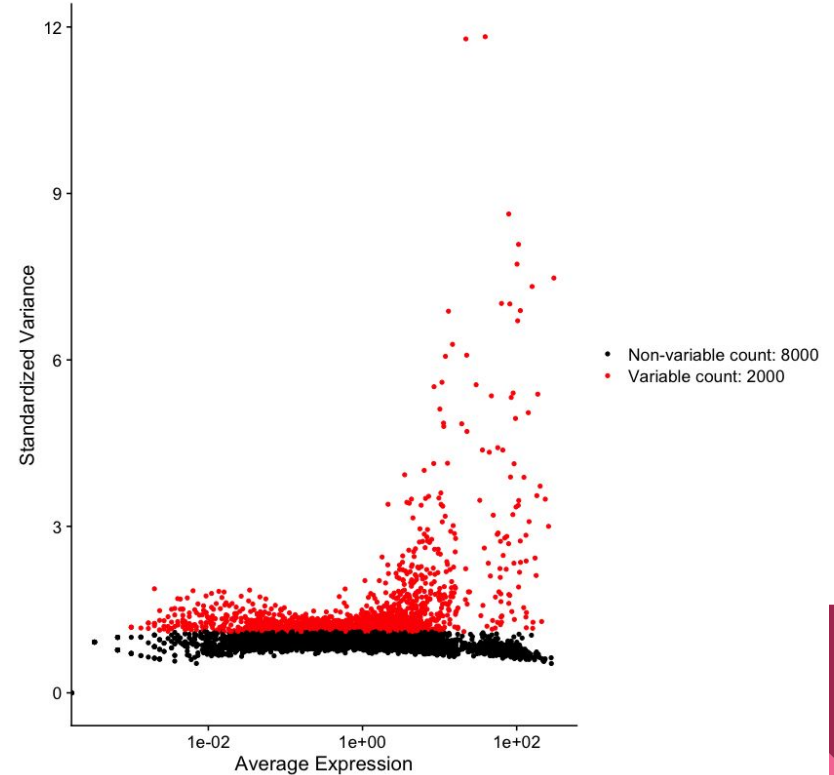
GRÁFICO DE CARACTERÍSTICAS VARIABLES

VariableFeaturePlot

Simulación 1



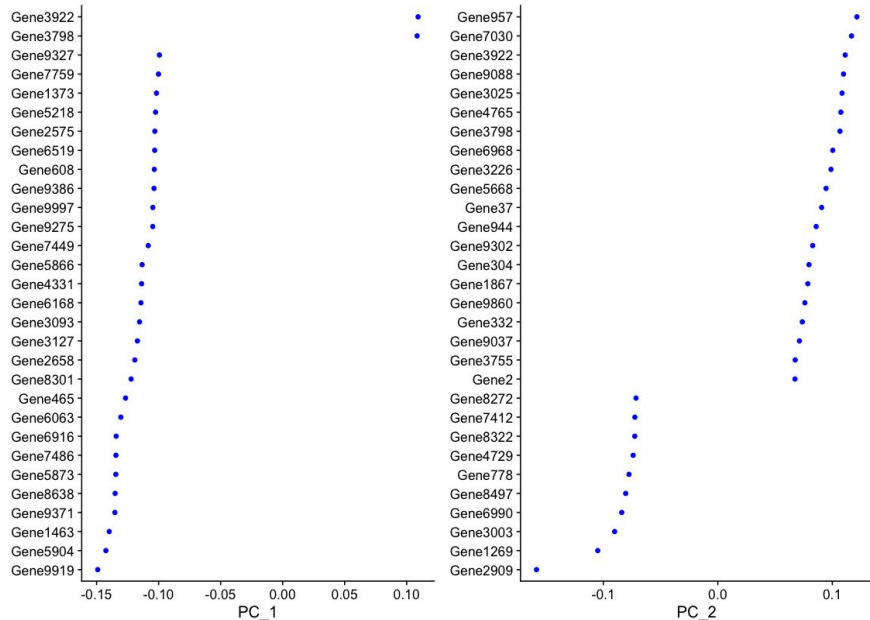
Simulación 2



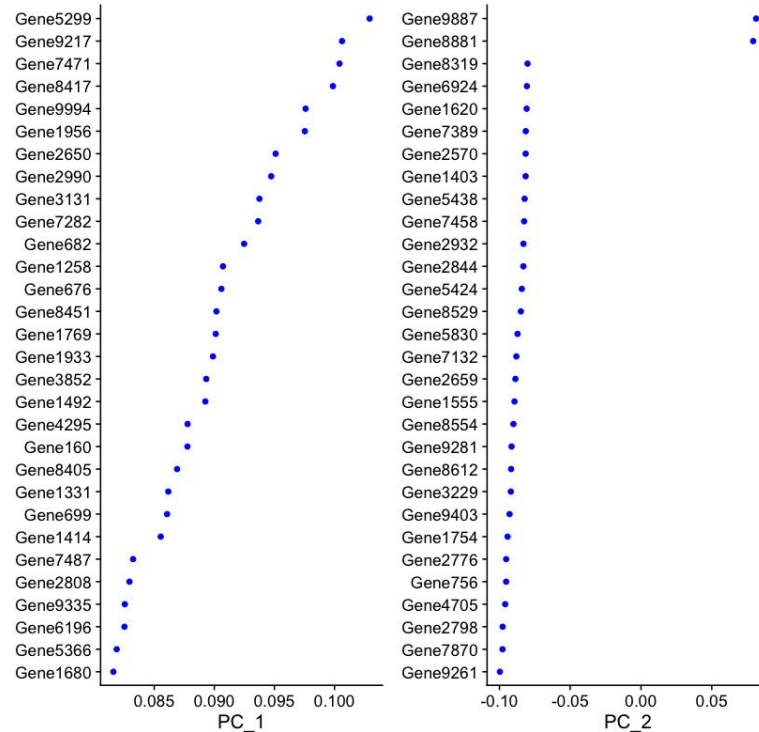
VISUALIZACIÓN DE LOS PESOS DE LOS COMPONENTES PRINCIPALES

VizDimLoadings

Simulación 1

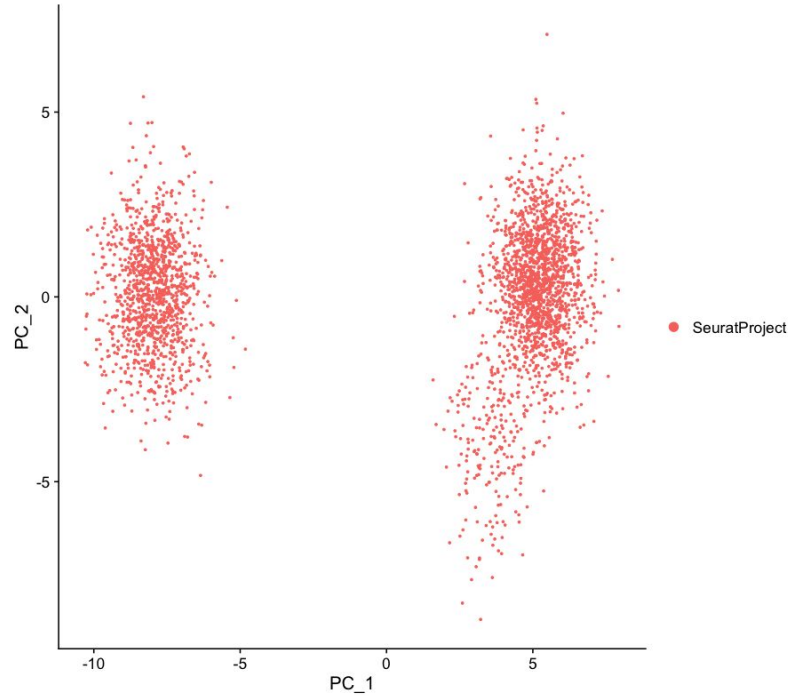


Simulación 2



REDUCCIÓN DE DIMENSIONALIDAD PCA

Simulación 1



Simulación 2

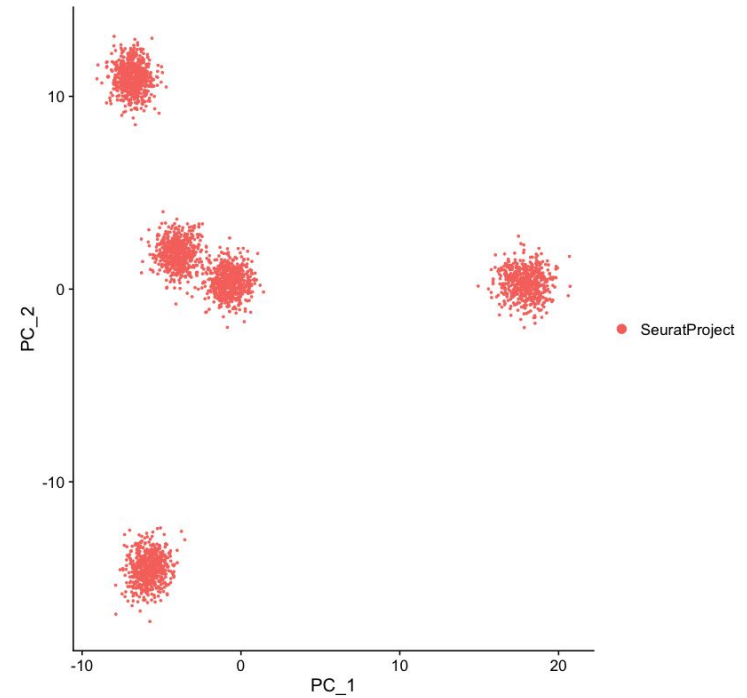
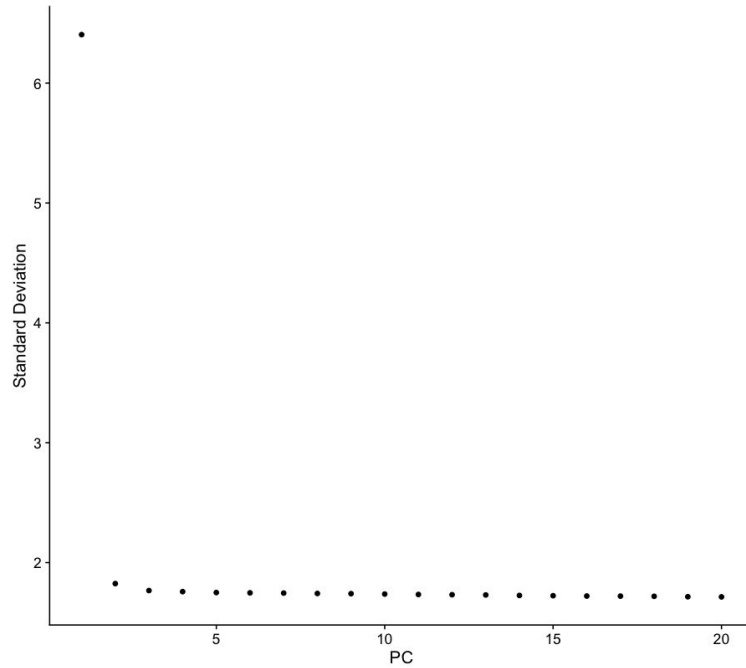
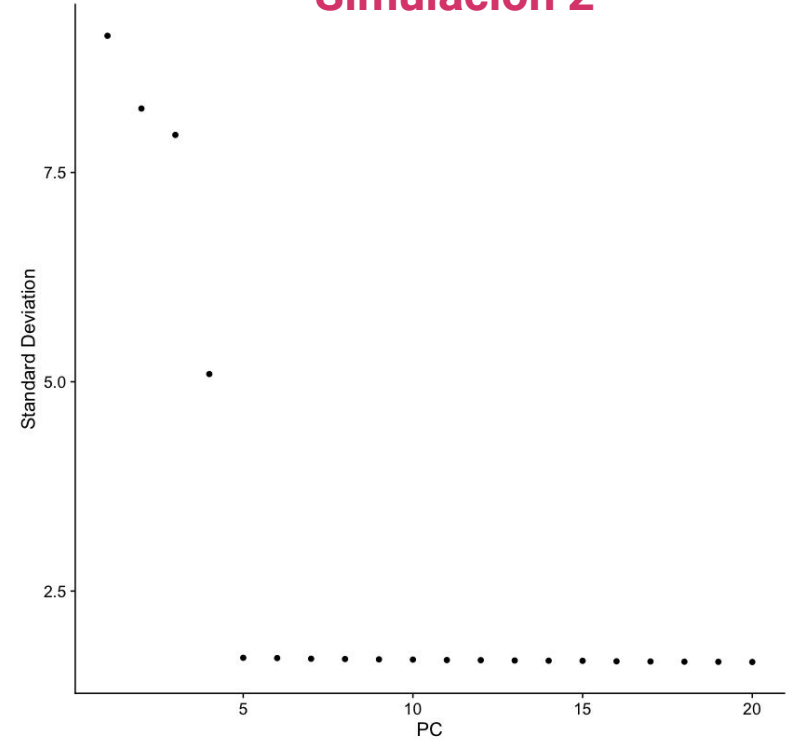


DIAGRAMA DE CODO (ELBOW PLOT)

Simulación 1

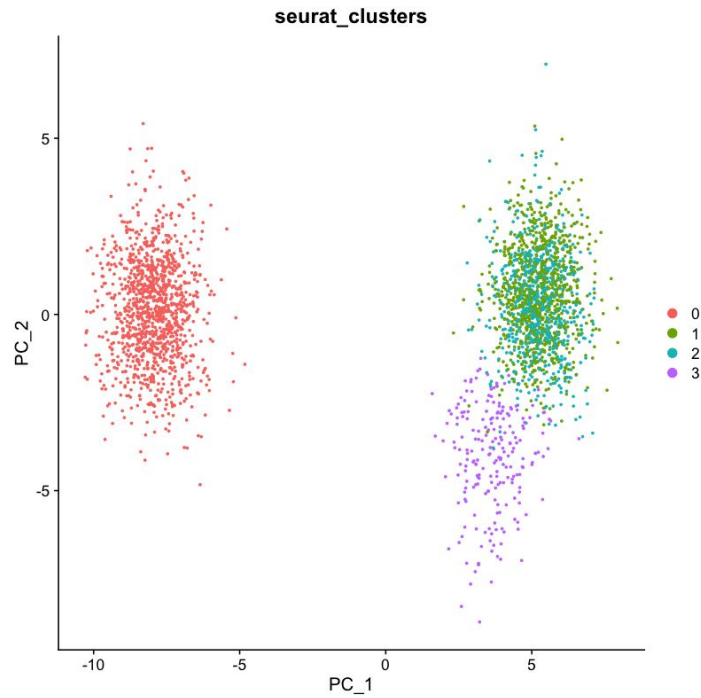


Simulación 2

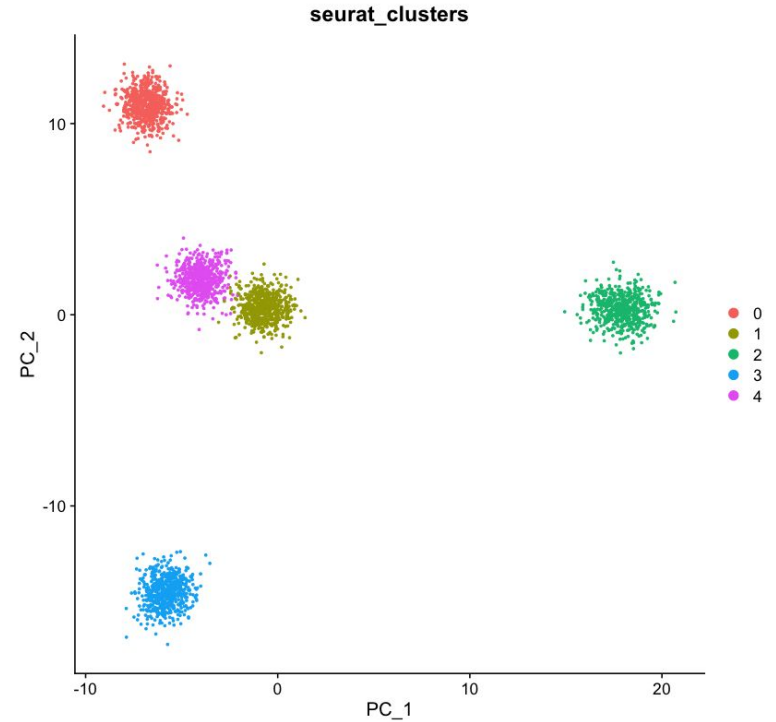


CLUSTERING

Simulación 1



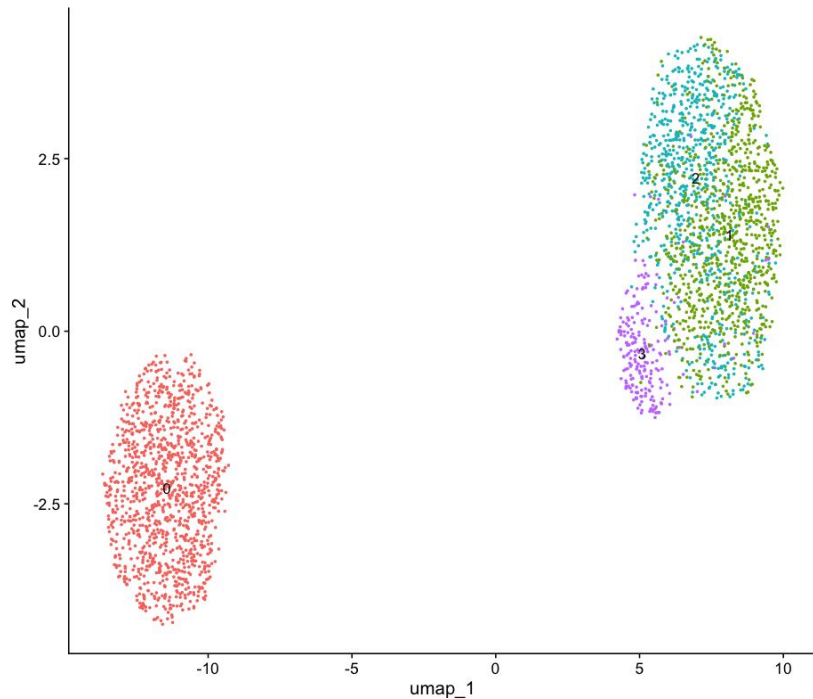
Simulación 2



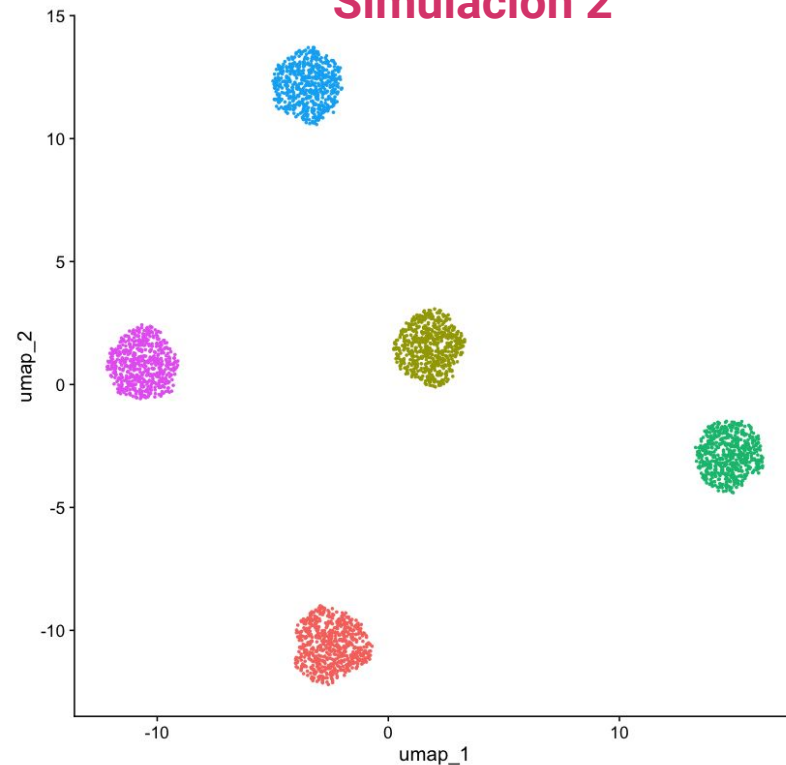
REDUCCIÓN DE DIMENSIONALIDAD

UMAP (Mapeo Uniforme de Manifold y Aproximación Probabilística)

Simulación 1



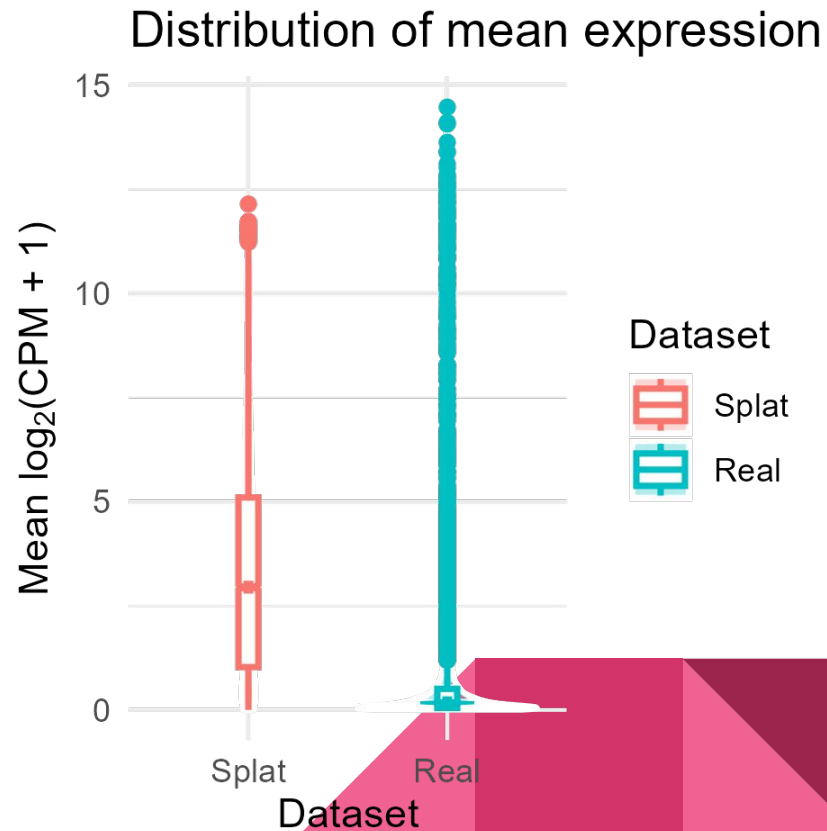
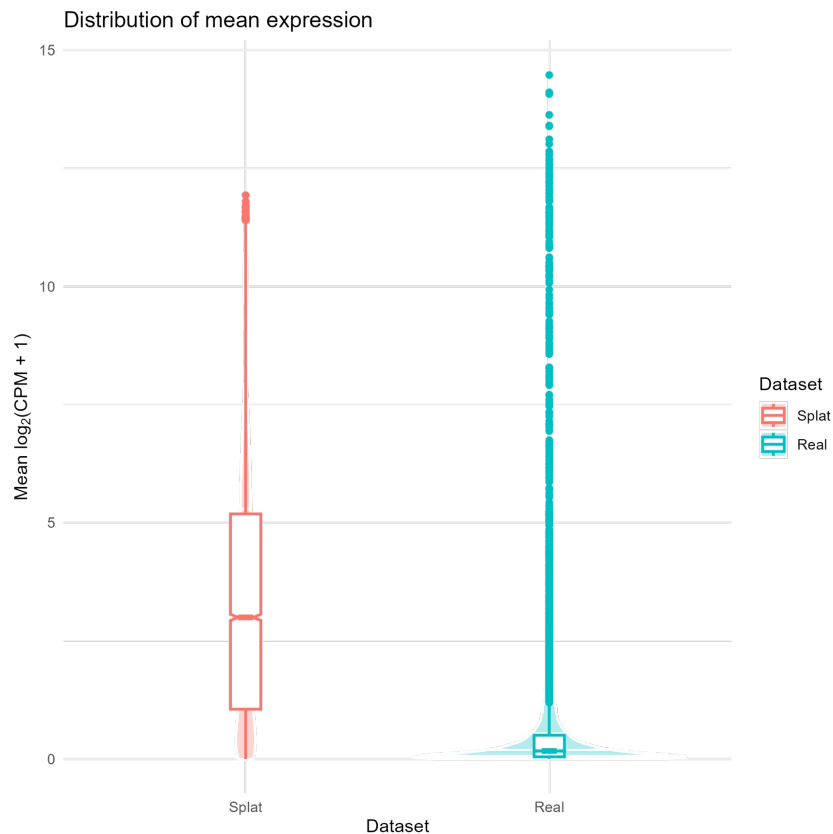
Simulación 2



```
group.prob = c(0.45, 0.23, 0.15, 0.10, 0.04, 0.03),
```

```
de.facscale = c(0.5, 0.5, 0.5, 0.5, 0.5, 0.5),
```

```
de.facscale = c(0.9, 0.9, 0.9, 0.9, 0.9, 0.9)
```



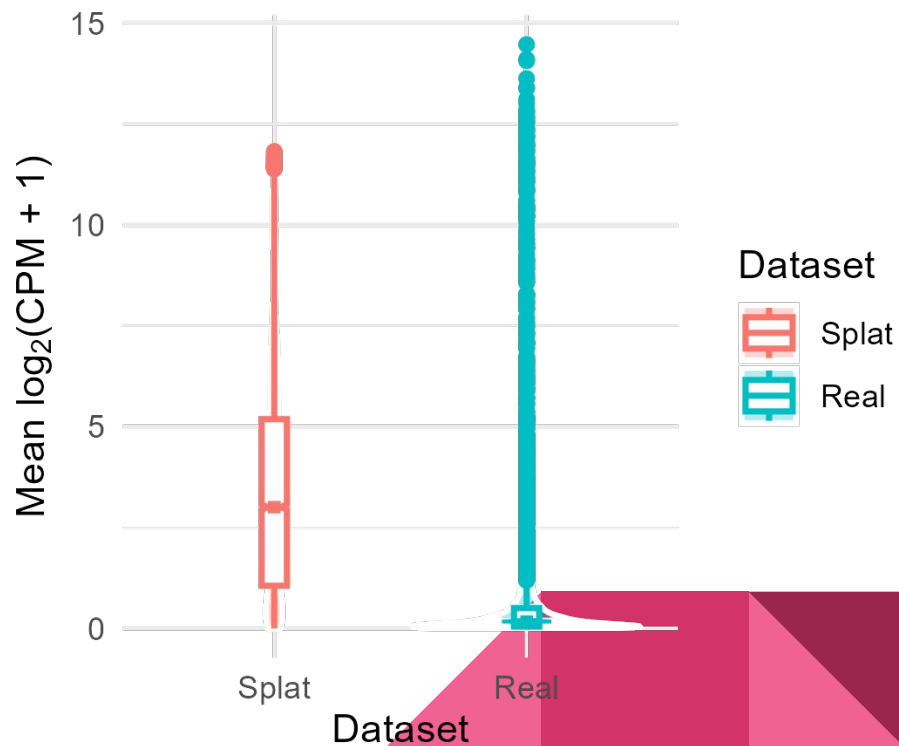
```
de.facscale = c(0.5, 0.8, 0.6, 0.9, 0.7, 0.8)
```

```
de.facscale = c(0.5, 0.2, 0.6, 0.2, 0.7, 0.2)
```

Distribution of mean expression

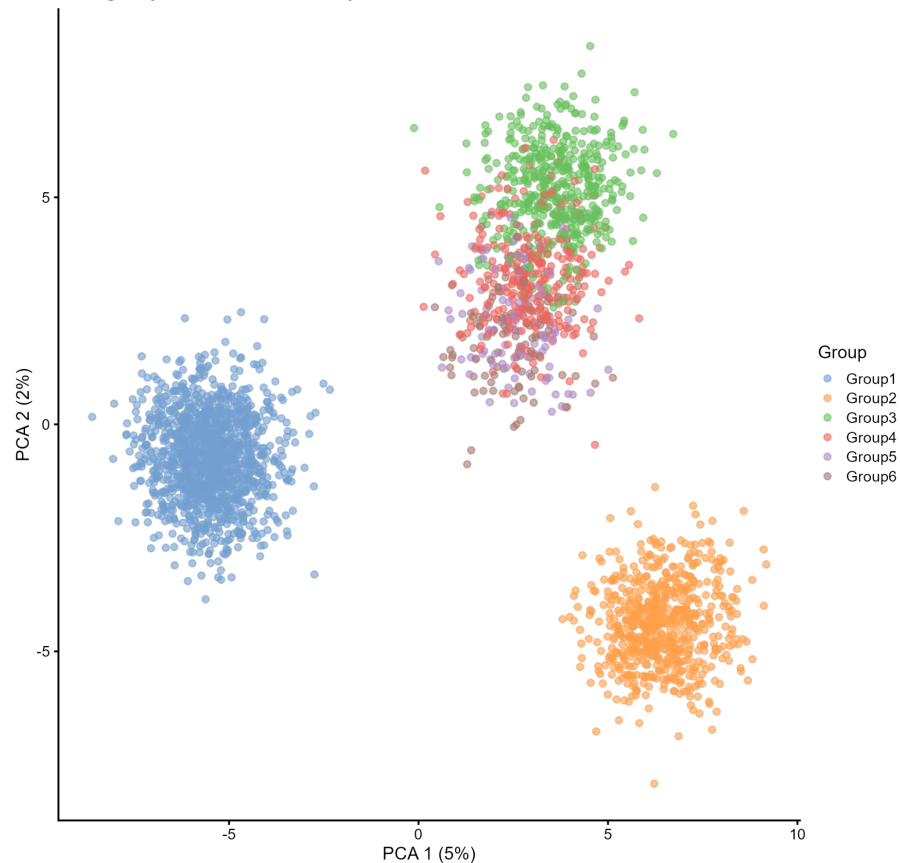


Distribution of mean expression



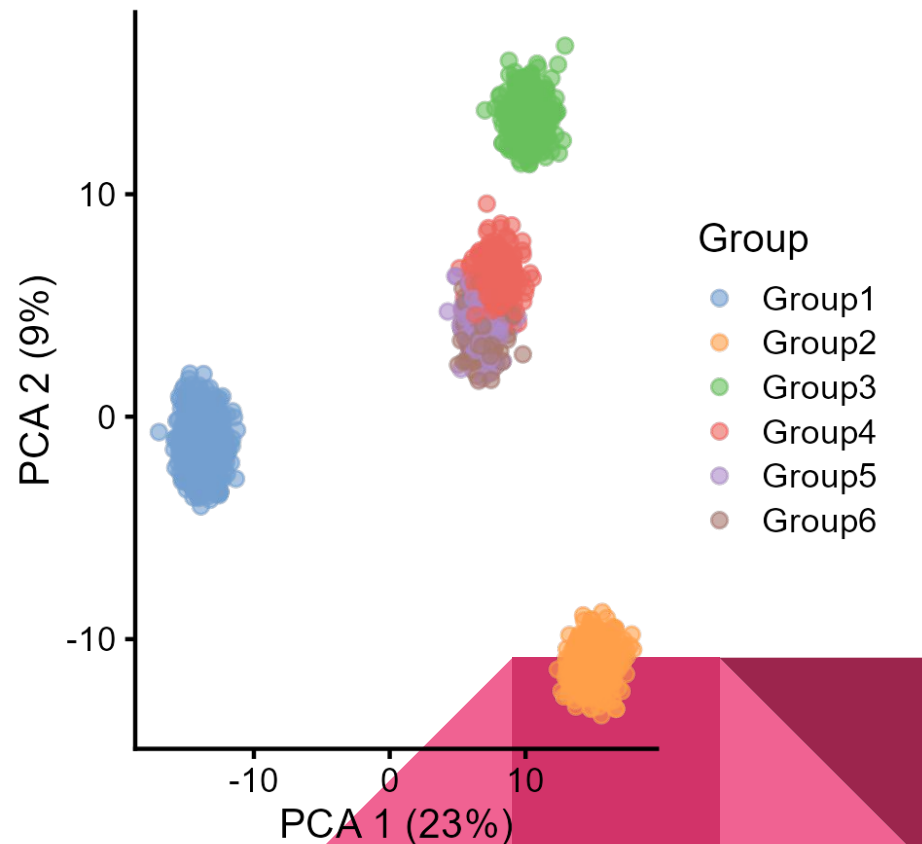
```
de.facscale = c(0.5, 0.5, 0.5, 0.5, 0.5, 0.5),
```

Five groups with different DE probabilities



```
de.facscale = c(0.9, 0.9, 0.9, 0.9, 0.9, 0.9)
```

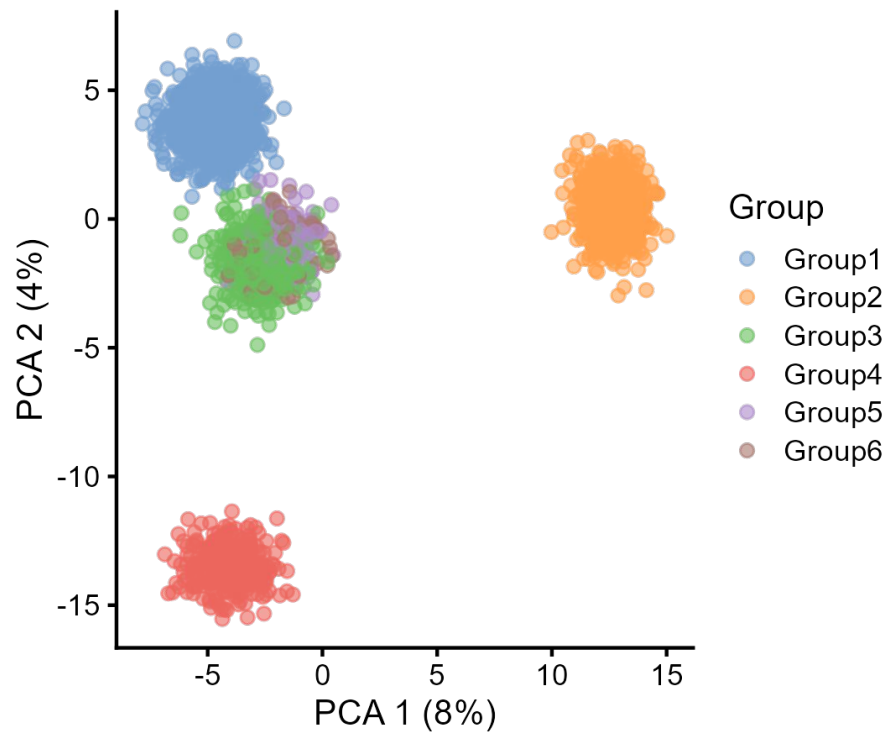
Five groups with different DE prob



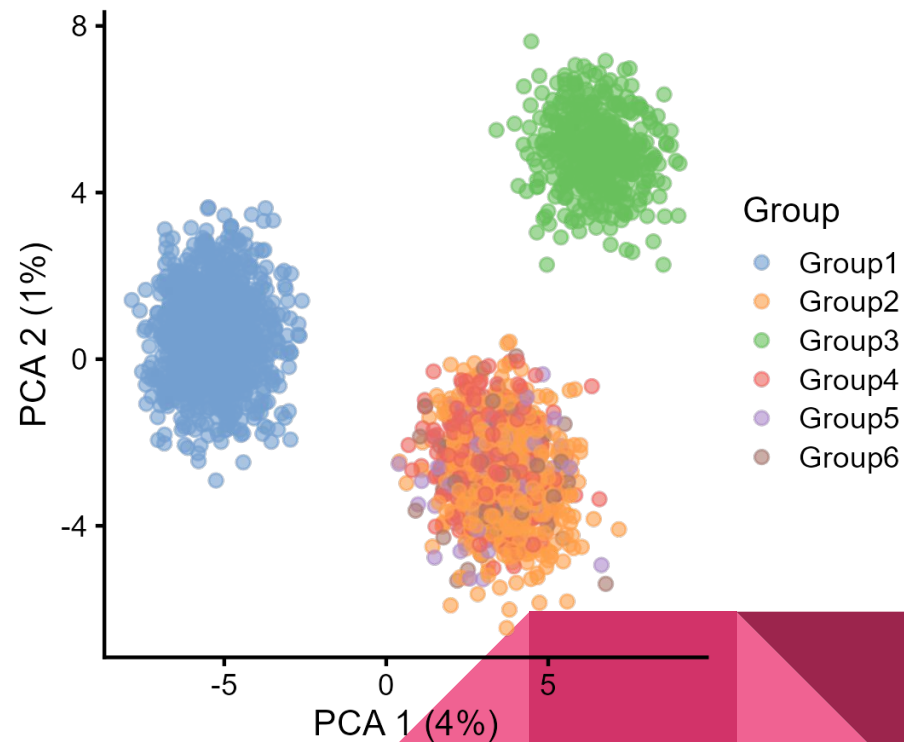
```
de.facscale = c(0.5, 0.8, 0.6, 0.9, 0.7, 0.8)
```

```
de.facscale = c(0.5, 0.2, 0.6, 0.2, 0.7, 0.2)
```

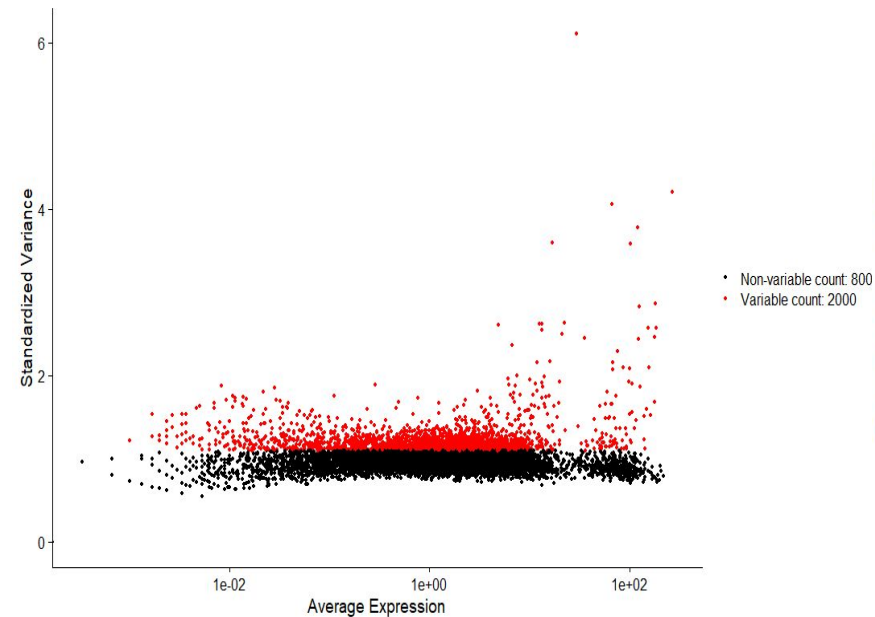
Five groups with different DE probabilities



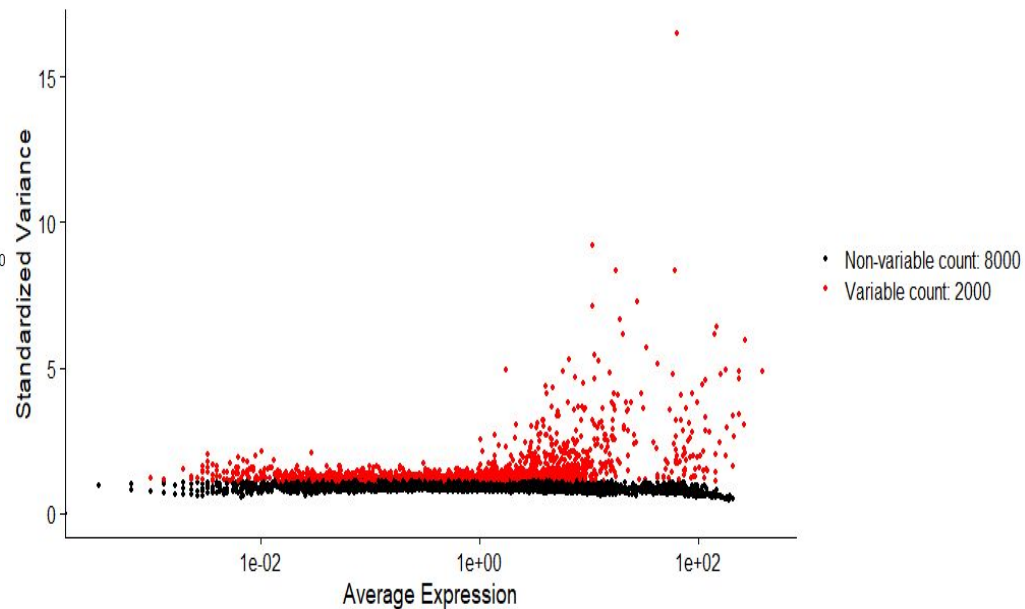
Five groups with different DE probabilities



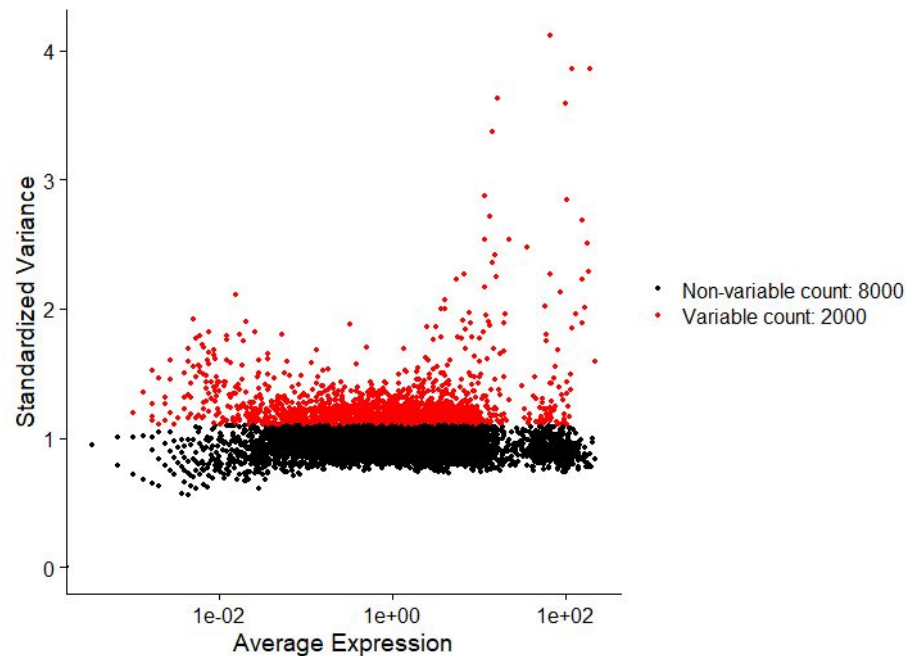
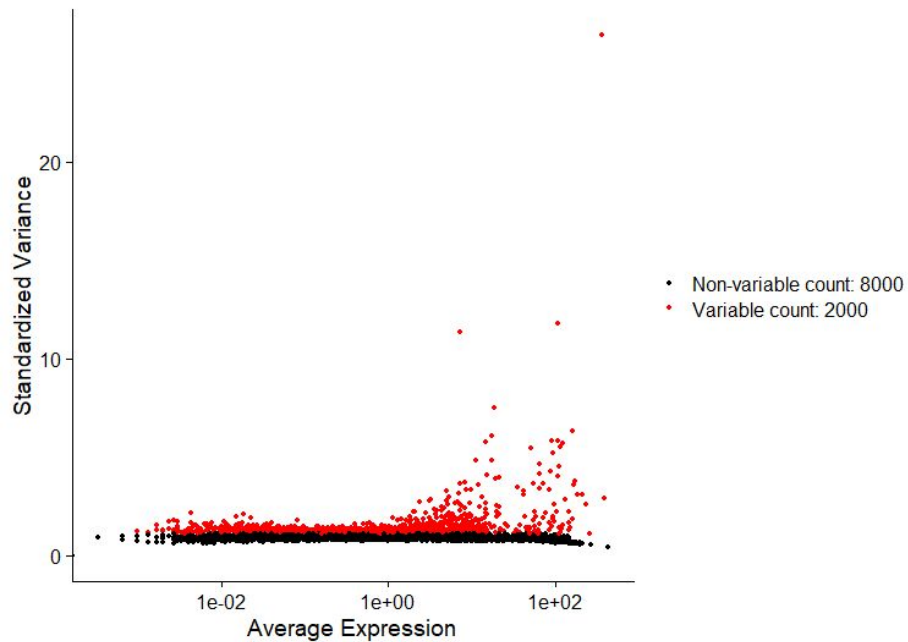
```
de.facscale = c(0.5, 0.5, 0.5, 0.5, 0.5, 0.5),
```



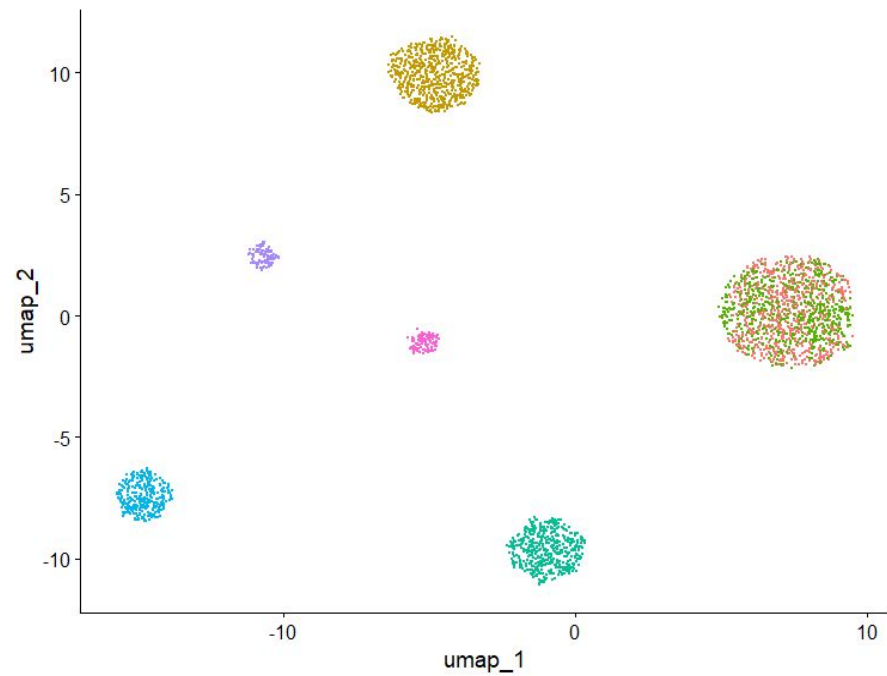
```
de.facscale = c(0.9, 0.9, 0.9, 0.9, 0.9, 0.9)
```



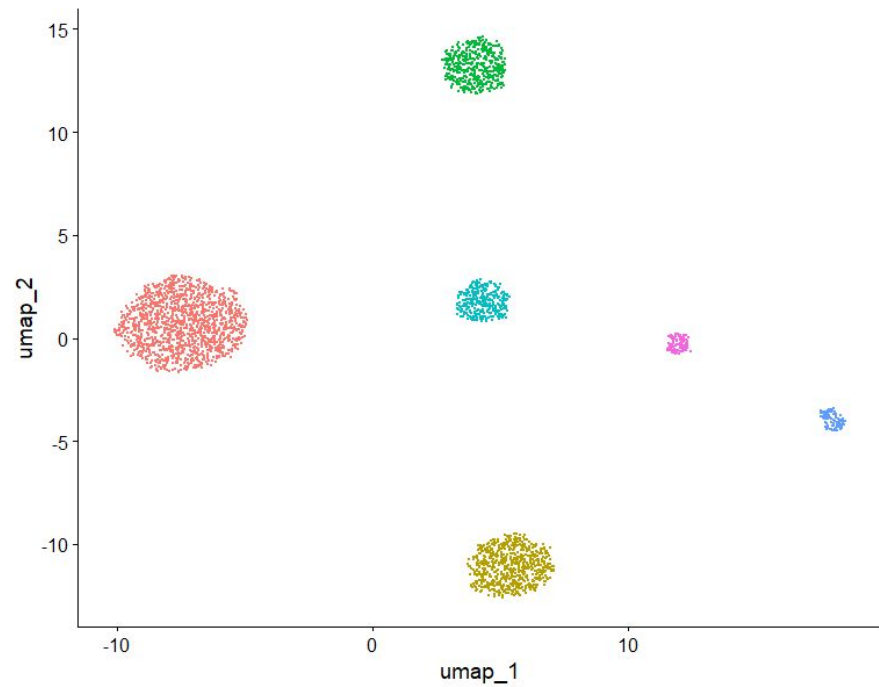
```
de.facscale = c(0.5, 0.8, 0.6, 0.9, 0.7, 0.8)  de.facscale = c(0.5, 0.2, 0.6, 0.2, 0.7, 0.2)
```



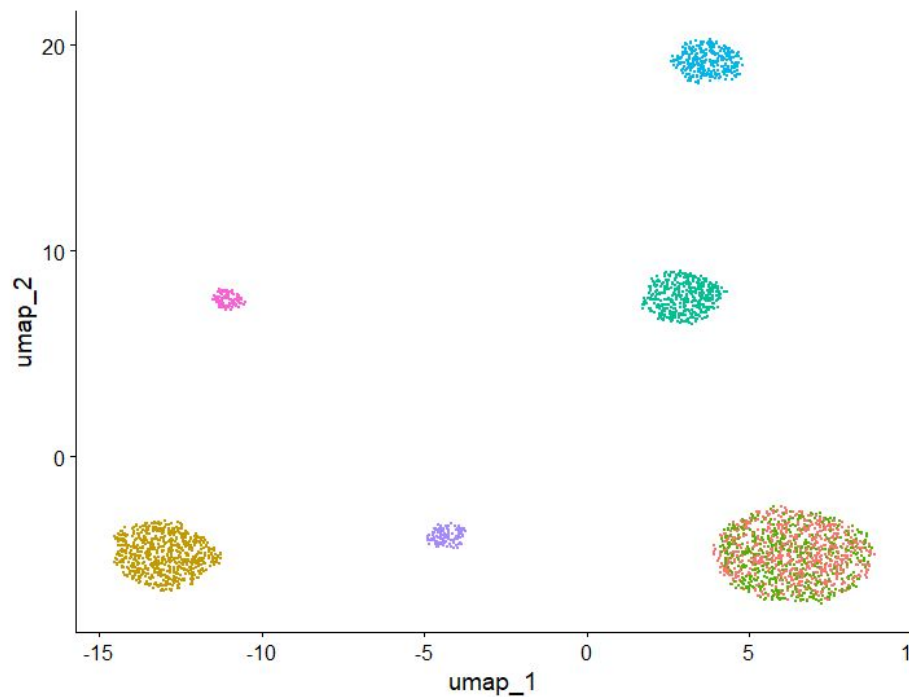
```
de.facscale = c(0.5, 0.5, 0.5, 0.5, 0.5, 0.5),
```



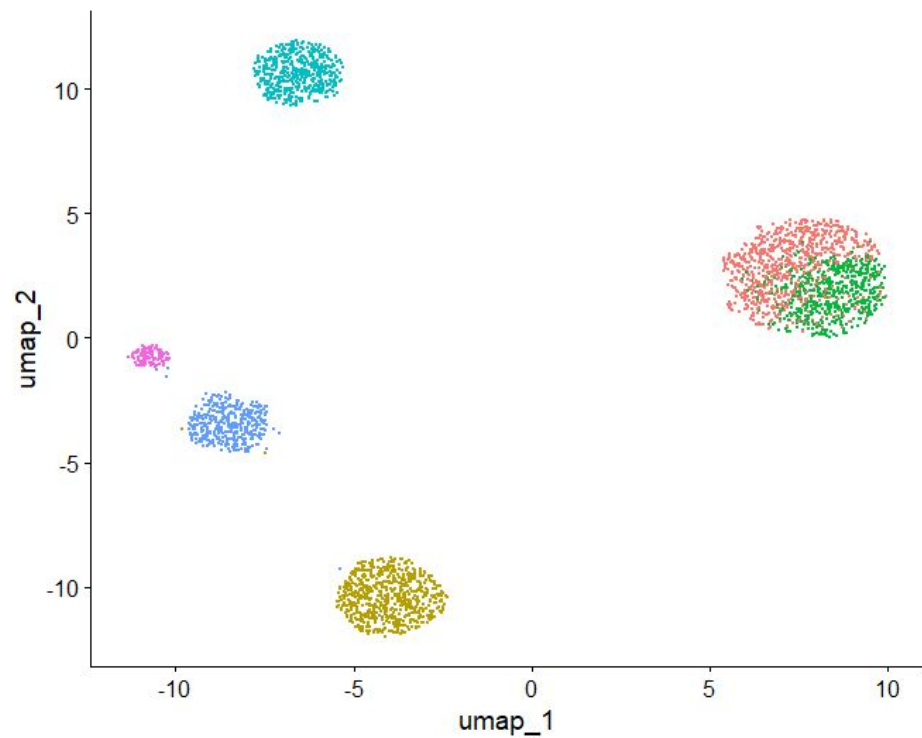
```
de.facscale = c(0.9, 0.9, 0.9, 0.9, 0.9, 0.9)
```




```
de.facscale = c(0.5, 0.8, 0.6, 0.9, 0.7, 0.8)
```



```
de.facscale = c(0.5, 0.2, 0.6, 0.2, 0.7, 0.2)
```



Bibliografía

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Nguyen, A., Khoo, W. H., Moran, I., Croucher, P. I., & Phan, T. G. (2018). Single Cell RNA Sequencing of Rare Immune Cell Populations. *Frontiers in immunology*, 9, 1553. <https://doi.org/10.3389/fimmu.2018.01553>

Reyfman, P. A., Walter, J. M., Joshi, N., Anekalla, K. R., McQuattie-Pimentel, A. C., Chiu, S., Fernandez, R., Akbarpour, M., Chen, C. I., Ren, Z., Verma, R., Abdala-Valencia, H., Nam, K., Chi, M., Han, S., Gonzalez-Gonzalez, F. J., Soberanes, S., Watanabe, S., Williams, K. J. N., Flozak, A. S., ... Misharin, A. V. (2019). Single-Cell Transcriptomic Analysis of Human Lung Provides Insights into the Pathobiology of Pulmonary Fibrosis. *American journal of respiratory and critical care medicine*, 199(12), 1517–1536. <https://doi.org/10.1164/rccm.201712-24100C>

Schirmer, L., Velmesshev, D., Holmqvist, S., Kaufmann, M., Werneburg, S., Jung, D., Vistnes, S., Stockley, J. H., Young, A., Steindel, M., Tung, B., Goyal, N., Bhaduri, A., Mayer, S., Engler, J. B., Bayraktar, O. A., Franklin, R. J. M., Haeussler, M., Reynolds, R., Schafer, D. P., ... Rowitch, D. H. (2019). Neuronal vulnerability and multilineage diversity in multiple sclerosis. *Nature*, 573(7772), 75–82. <https://doi.org/10.1038/s41586-019-1404-z>

Svensson, V., da Veiga Beltrame, E., & Pachter, L. (2020). A curated database reveals trends in single-cell transcriptomics. *Database : the journal of biological databases and curation*, 2020, baaa073. <https://doi.org/10.1093/database/baaa073>

```

set.seed(42)

# de.facScale variado > 0.5
sim.groups <- splatSimulate(
  params,
  group.prob = c(0.45, 0.23, 0.15, 0.10, 0.04, 0.03),
  de.prob = c(0.08, 0.06, 0.05, 0.05, 0.01, 0.02),
  de.facLoc = c(0.5, 1, 0.8, 0.8, 0.5, 1),
  de.facScale = c(0.7, 0.8, 0.6, 0.9, 0.7, 0.8),
  batchCells=3000,
  method = "groups",
  verbose = TRUE
)

# de.facScale variado = 0.9
sim.groups_2 <- splatSimulate(
  params,
  group.prob = c(0.45, 0.23, 0.15, 0.10, 0.04, 0.03),
  de.prob = c(0.08, 0.06, 0.05, 0.05, 0.01, 0.02),
  de.facLoc = c(0.5, 1, 0.8, 0.8, 0.5, 1),
  de.facScale = c(0.9),
  batchCells=3000,
  method = "groups",
  verbose = TRUE
)

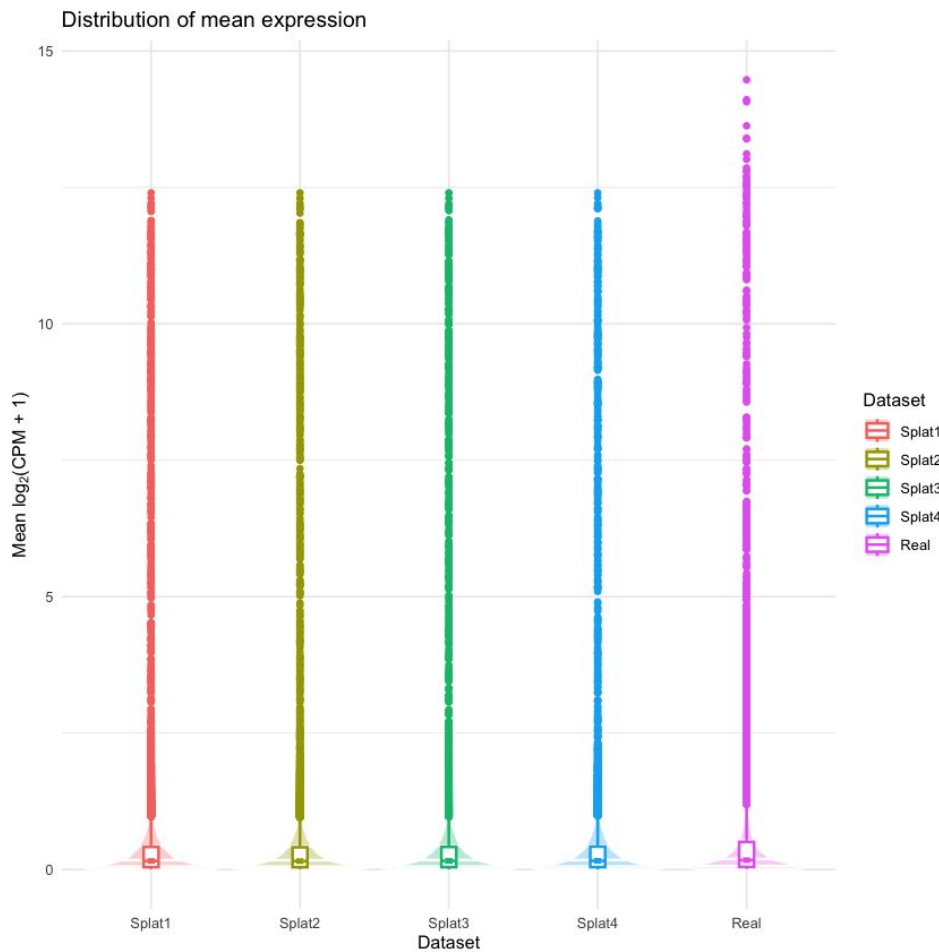
```

```

# de.facScale variado = 0.5
sim.groups_3 <- splatSimulate(
  params,
  group.prob = c(0.45, 0.23, 0.15, 0.10, 0.04, 0.03),
  de.prob = c(0.08, 0.06, 0.05, 0.05, 0.01, 0.02),
  de.facLoc = c(0.5, 1, 0.8, 0.8, 0.5, 1),
  de.facScale = c(0.5),
  batchCells=3000,
  method = "groups",
  verbose = TRUE
)

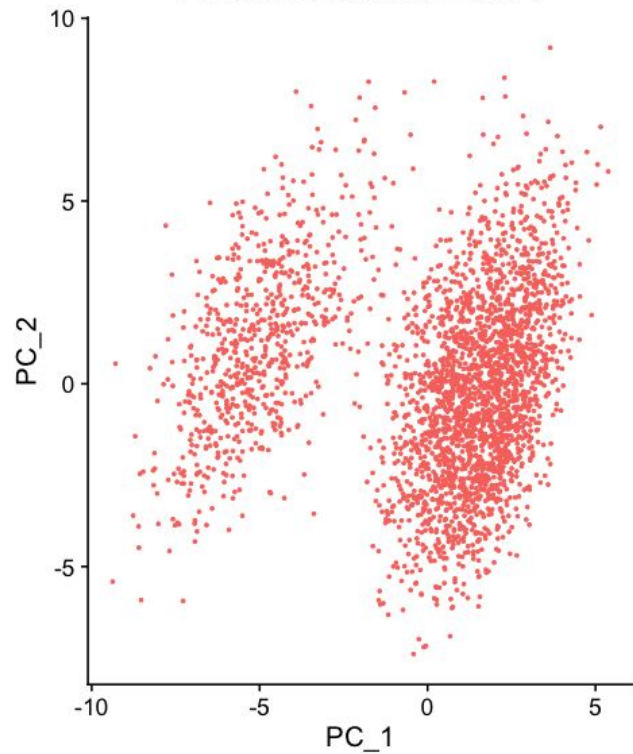
# de.facScale variado < 0.5
sim.groups_4 <- splatSimulate(
  params,
  group.prob = c(0.45, 0.23, 0.15, 0.10, 0.04, 0.03),
  de.prob = c(0.08, 0.06, 0.05, 0.05, 0.01, 0.02),
  de.facLoc = c(0.5, 1, 0.8, 0.8, 0.5, 1),
  de.facScale = c(0.2, 0.5, 0.2, 0.5, 0.4, 0.3),
  batchCells=3000,
  method = "groups",
  verbose = TRUE
)

```



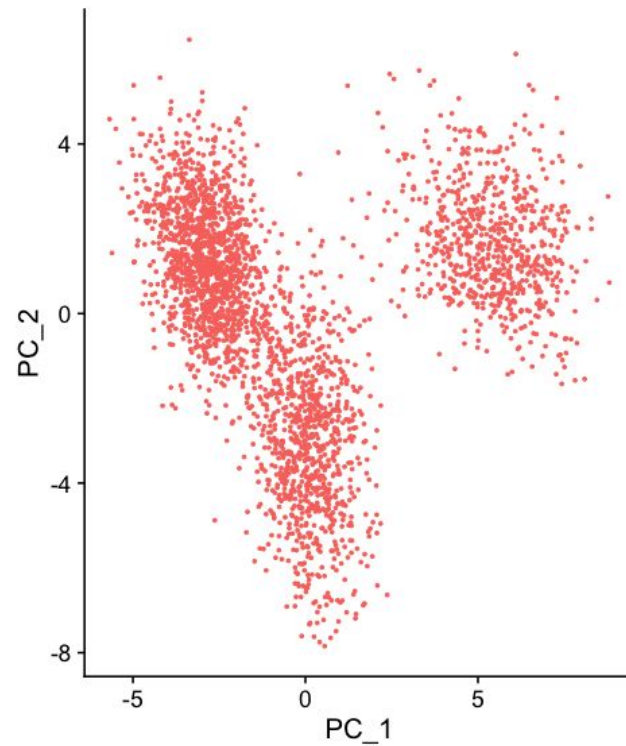
≠ Visualizar la comparación

PCA of Simulated Data 1



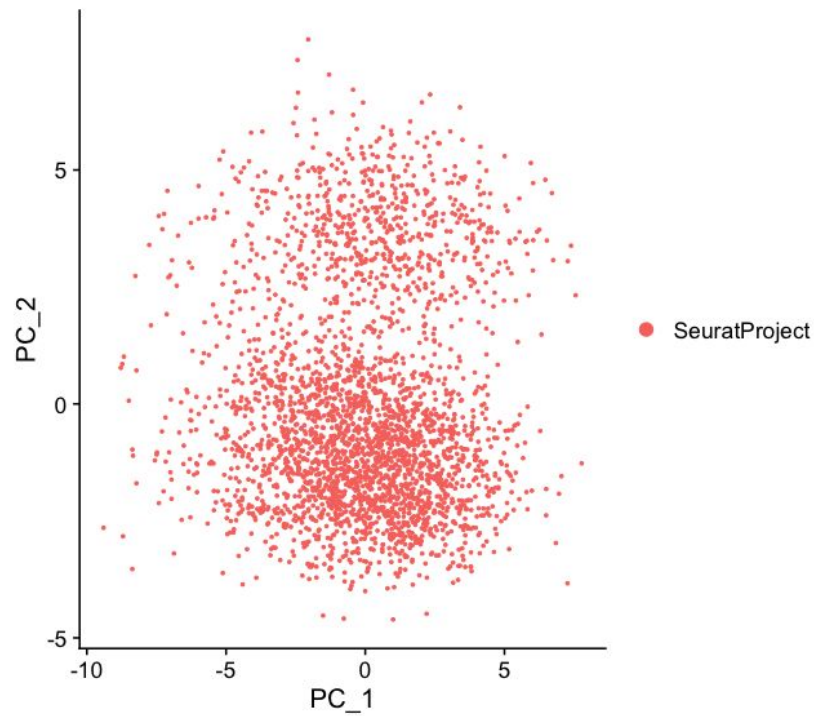
● SeuratProject

PCA of Simulated Data 2

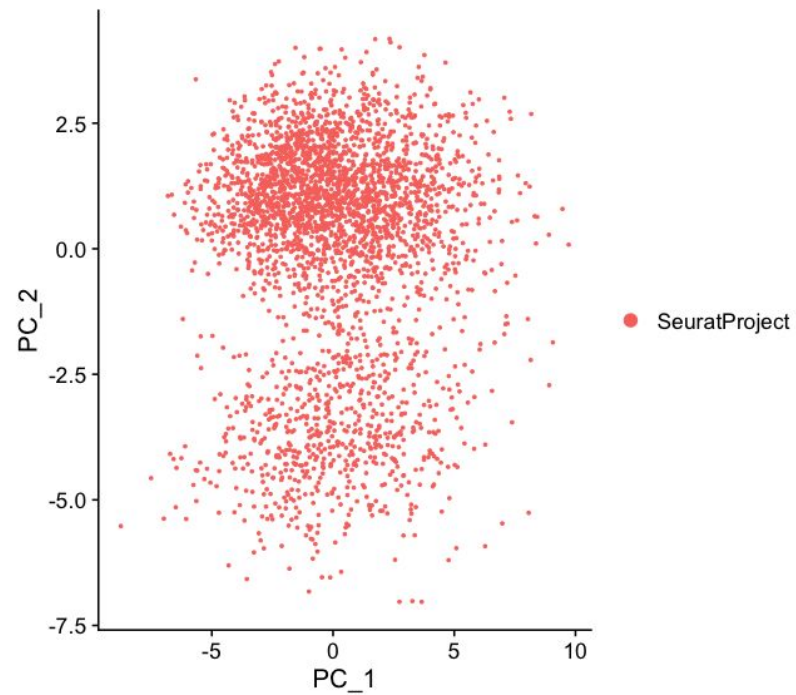


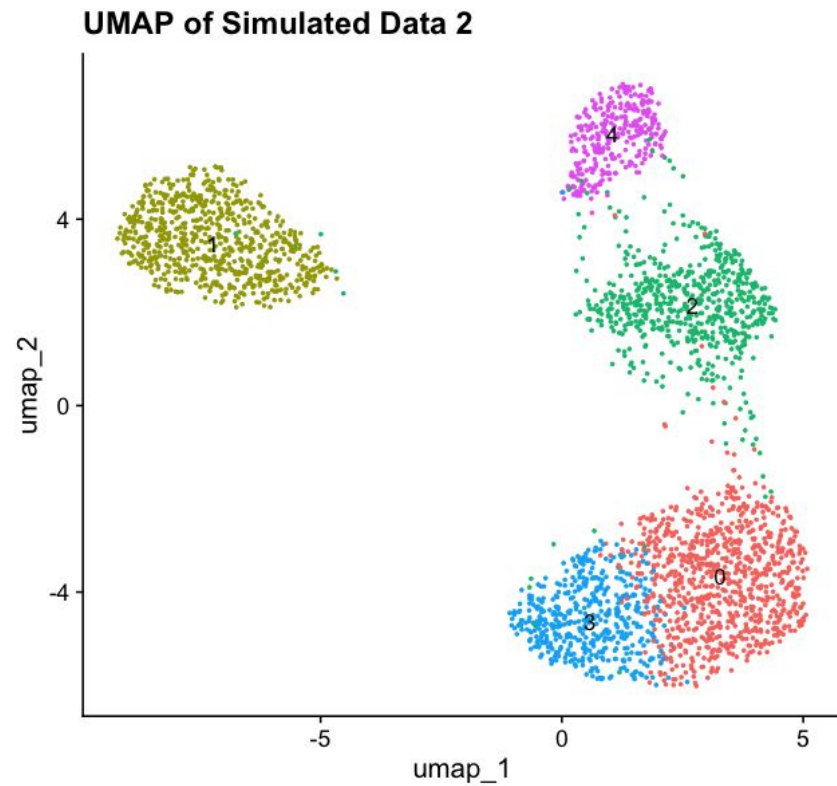
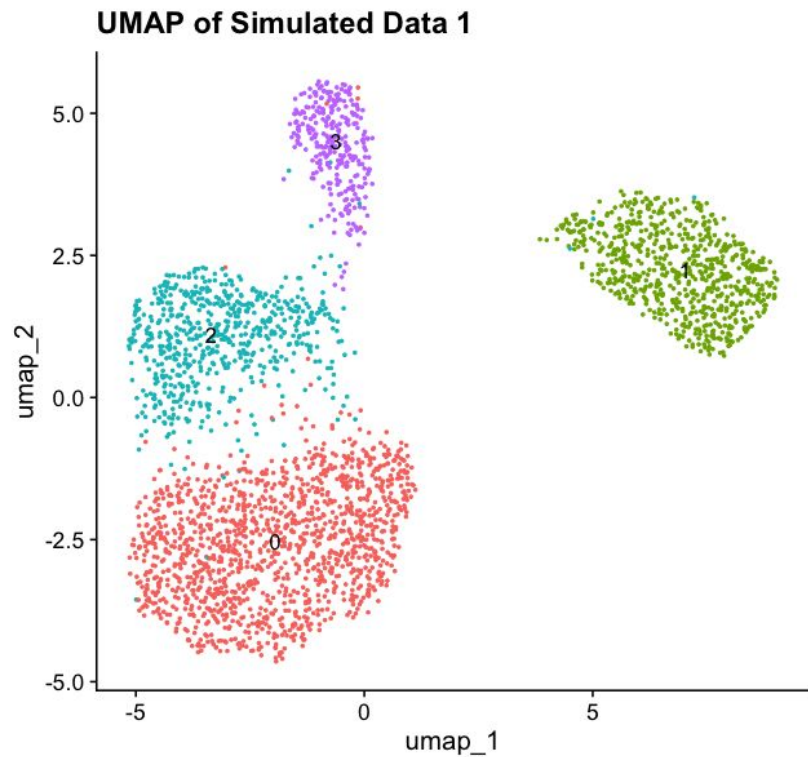
● SeuratProject

PCA of Simulated Data 3

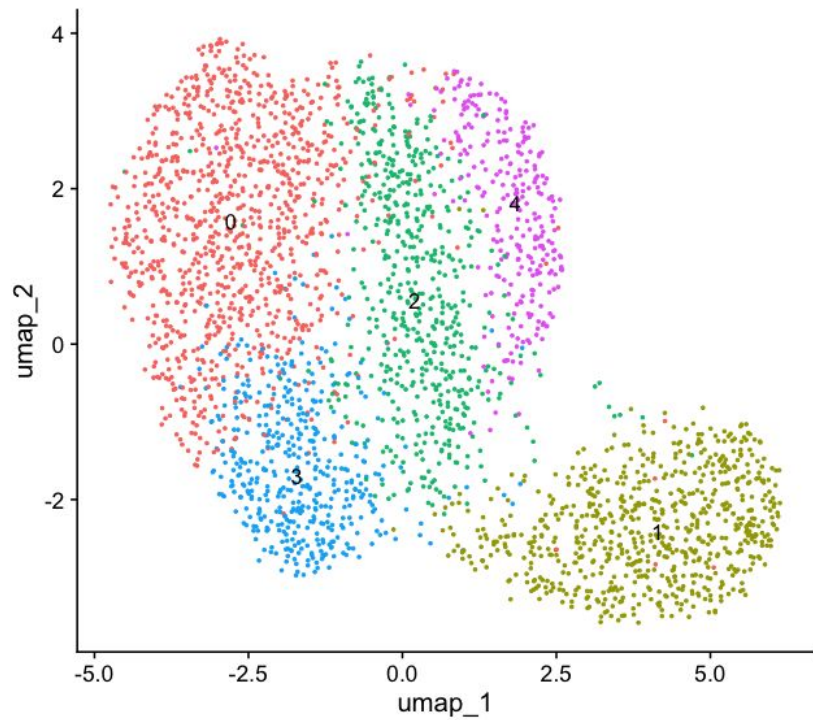


PCA of Simulated Data 4

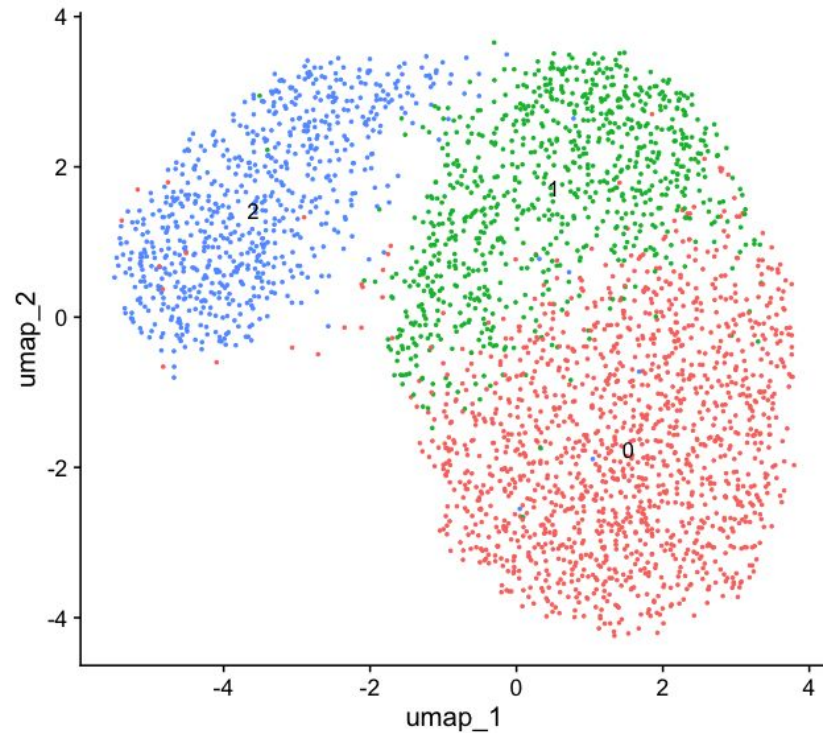




UMAP of Simulated Data 3



UMAP of Simulated Data 4



Distribution of variance

